

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD2-090***

***Handsom - Southampton 115 kV
42 MW Capacity / 70 MW Energy***

December / 2019

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between Invenergy Solar Development North America LLC, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Virginia Electric and Power Company (VEPCO).

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the IC. As a requirement for interconnection, the IC may be responsible for the cost of constructing Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an IC may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The IC is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

The IC has proposed a solar generating facility located in Southampton County, Virginia. The installed facilities will have a total capability of 70 MW with 42 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is 12/31/2020. This study does not imply an ITO commitment to this in-service date.

Point of Interconnection

AD2-090 will interconnect with the ITO transmission system via a tap off of the Handsom - Southampton 115 kV line. The primary Point of Interconnection (POI) chosen was a tap off of the Handsom - Southampton 115 kV line through an interconnection switchyard 3 miles from Southampton Substation and 2 miles from Handsom Substation. See one line in Attachment 1.

Cost Summary

The AD2-090 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 1,550,000
Direct Connection Network Upgrades	\$ 5,500,000
Non Direct Connection Network Upgrades	\$ 800,000
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 6,814,200
Total Costs	\$ 14,664,200

Attachment Facilities

Generation Substation: Install metering and associated protection equipment. Estimated Cost \$550,000.

Transmission: Construct approximately one span of 115 kV Attachment line between the generation substation and a new AD2-090 Switching Station. The estimated cost for this work is \$1,000,000.

The estimated total cost of the Attachment Facilities is \$1,550,000. It is estimated to take 18-24 months to complete this work. These preliminary cost estimates are based on typical engineering costs. A more detailed engineering cost estimates are normally done when the IC provides an exact site plan location for the generation substation during the Facility Study phase. See Attachment One.

Direct Connection Cost Estimate

Substation: Establish the new 115 kV AD2-090 Switching Substation (interconnection substation). The arrangement of the substation will be as shown below in Attachment 1: One-Line Diagram. The estimated cost of this work scope is \$5,500,000. It is estimated to take 24-36 months to complete this work.

Non-Direct Connection Cost Estimate

Transmission: Install transmission structure in-line with transmission line to allow the proposed interconnection switching station to be interconnected with the transmission system. Estimated cost is \$800,000 and it is estimated to take 24-30 months to complete. See Attachment 1.

Remote Terminal Work: During the Facilities Study, ITO's System Protection Engineering Department will review transmission line protection as well as anti-islanding required to accommodate the new generation and interconnection substation. System Protection Engineering will determine the minimal acceptable protection requirements to reliably interconnect the proposed generating facility with the transmission system. The review is based on maintaining system reliability by reviewing ITO's protection requirements with the known transmission system configuration which includes generating facilities in the area. This review may determine that transmission line protection and communication upgrades are required at remote substations.

New System Reinforcements

PJM OATT 217.3 outlines cost responsibility for Network Upgrades and as the minimum amount of Network Upgrades required to resolve a single reliability criteria violation will not meet or exceed \$5,000,000 such costs shall be allocated to those Interconnection Requests in the New Services Queue that contribute to the need for such upgrades. Such allocations shall be made in proportion to each Interconnection Request's megawatt contribution to the need for these upgrades subject to the rules for minimum cost allocation thresholds in the PJM Manuals. For the purpose of applying the \$5,000,000 threshold, each reliability criteria violation shall be considered separately.

Violation #	Ruling Violation #	Loading	Upgrade Description	Upgrade Cost	Allocated Cost
# 1	1	From 98.66% To 100.61%	Wreck and rebuild 0.98 miles of the Chestnut - Cox DP 115 kV line	\$1,960,000	\$0
# 2, 16	2	From 99.29% To 100.48%	Rebuild Clubhouse-Lakeview 230 kV Line #254	\$27,000,000	\$0
# 3, 4	3	From 131.78% To 133.71%	Wreck and rebuild 4.39 miles of the Chestnut – Whitakers 115 kV line #1001	\$8,780,000	\$0
# 5	5	From 154.38% To 155.71%	Rebuild the 4.3 miles of Dominion 230 kV Line #2058 Rocky Mt. – Hathaway	\$13,000,000	\$0
# 6	6	From 261.68% To 264.14%	Battleboro – Rocky Mt. 115 kV line (Dominion – Duke Energy/Progress tie line)		\$0
# 7, 8, 9	7	From 120.91% To 125.2%	Add a second 230-115 kV transformer at the Carolina substation	\$5,500,000	\$824,703
# 10, 11	10	From 119.67% To 120.92%	Replace wave trap at Clubhouse Substation. Rating: 722 MVA	\$150,000	\$0
# 12	12	From 102.3% To 106.91%	Add a second 230-115 kV transformer at the Earleys substation	\$5,500,000	\$4,720,677
# 14, 15	14	From 132.75% To 134.68%	Wreck and rebuild 4.55miles of the Whitakers– Battleboro 115kV line # 1001	\$9,100,000	\$0
# 13, 17	17	From 140.08% To 141.23%	Rebuild the 20.5 miles of Dominion 230 kV Line #218 Everetts - Greenville	\$30,750,000	\$1,268,820
Total Estimate Allocated Cost of Network Upgrades					\$6,814,200

Interconnection Customer Requirements

ITO's Facility Interconnection Requirements as posted on PJM's website

<http://www.pjm.com/~media/planning/plan-standards/private-dominion/facility-connection-requirements1.ashx>

Voltage Ride Through Requirements - The Customer Facility shall be designed to remain in service (not trip) for voltages and times as specified for the Eastern Interconnection in Attachment 1 of NERC Reliability Standard PRC-024-1, and successor Reliability Standards, for both high and low voltage conditions, irrespective of generator size, subject to the permissive trip exceptions established in PRC-024-1 (and successor Reliability Standards).

Frequency Ride Through Requirements - The Customer Facility shall be designed to remain in service (not trip) for frequencies and times as specified in Attachment 2 of NERC Reliability Standard PRC-024-1, and successor Reliability Standards, for both high and low frequency condition, irrespective of generator size, subject to the permissive trip exceptions established in PRC-024-1 (and successor Reliability Standards).

Reactive Power - The Generation Interconnection Customer shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading to 0.95 lagging measured at the generator's terminals.

Meteorological Data Reporting Requirement - The solar generation facility shall, at a minimum, be required to provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Interconnected Transmission Owner Requirements

Metering and SCADA/Communication equipment must meet the requirements outlined in section 3.1.6 Metering and Telecommunications of ITO's Facility Connection Requirement NERC Standard FAC-001 which is publically available at www.dom.com.

Network Impacts

The Queue Project AD2-090 was evaluated as a 70.0 MW (Capacity 42.0 MW) injection into Handsom - Southampton 115kV substation. Project AD2-090 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-090 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Contingency Descriptions

The following contingencies resulted in overloads:

Contingency Name	Description
DVP_P1-2: LN 130-A	CONTINGENCY 'DVP_P1-2: LN 130-A' OPEN BRANCH FROM BUS 314562 TO BUS 314570 CKT 1 /* 3CLUBHSE 115.00 - 3METCATP 115.00 OPEN BRANCH FROM BUS 314570 TO BUS 314572 CKT 1 /* 3METCATP 115.00 - 3EMPORIA 115.00 OPEN BRANCH FROM BUS 314570 TO BUS 314588 CKT 1 /* 3METCATP 115.00 - 3METCALF 115.00 OPEN BRANCH FROM BUS 314572 TO BUS 925170 CKT 1 /* 3EMPORIA 115.00 - AB2-174 TAP 115.00 OPEN BRANCH FROM BUS 314572 TO BUS 314863 CKT 1 /* 3EMPORIA 115.00 - 3EMPOR_1 115.00 OPEN BUS 314570 /* ISLAND OPEN BUS 314572 /* ISLAND OPEN BUS 314588 /* ISLAND END
DVP_P1-2: LN 140-C	CONTINGENCY 'DVP_P1-2: LN 140-C' OPEN BRANCH FROM BUS 920040 TO BUS 314558 CKT 1 /* AA2-088 TAP 115.00 - 3BOYKINS 115.00 END
DVP_P1-2: LN 2012-B	CONTINGENCY 'DVP_P1-2: LN 2012-B' OPEN BRANCH FROM BUS 936400 TO BUS 314569 CKT 1 /* AD2-051 TAP230.00 - 6EARLEYS 230.00 END
DVP_P1-2: LN 2056-A	CONTINGENCY 'DVP_P1-2: LN 2056-A' OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1 /* 6HATHAWAY 230.00 - AD1-057 TAP 230.00 END
DVP_P1-2: LN 2056-B	CONTINGENCY 'DVP_P1-2: LN 2056-B' OPEN BRANCH FROM BUS 934330 TO BUS 314579 CKT 1 /* AD1-057 TAP 230.00 - 6HORNRTN 230.00 END
DVP_P1-2: LN 2058	CONTINGENCY 'DVP_P1-2: LN 2058' OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6MORNSTR 230.00 END

DVP_P1-2: LN 2131A	CONTINGENCY 'DVP_P1-2: LN 2131A' OPEN BRANCH FROM BUS 314662 TO BUS 916040 CKT 1 /* 6S HERTFORD 230.00 - Z1-036 TAP 230.00 OPEN BRANCH FROM BUS 314651 TO BUS 314662 CKT 1 /* 6WINFALL 230.00 - 6S HERTFORD 230.00 OPEN BUS 314662 /* ISLAND END
DVP_P1-2: LN 2181	CONTINGENCY 'DVP_P1-2: LN 2181' OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END
DVP_P1-2: LN 238	CONTINGENCY 'DVP_P1-2: LN 238' OPEN BRANCH FROM BUS 314282 TO BUS 314435 CKT 1 /* 6CARSON 230.00 - 6SAPONY 230.00 OPEN BRANCH FROM BUS 314435 TO BUS 314563 CKT 1 /* 6SAPONY 230.00 - 6CLUBHSE 230.00 OPEN BRANCH FROM BUS 314562 TO BUS 314563 CKT 1 /* 3CLUBHSE 115.00 - 6CLUBHSE 230.00 OPEN BUS 314435 /* ISLAND END
DVP_P1-2: LN 246	CONTINGENCY 'DVP_P1-2: LN 246' OPEN BRANCH FROM BUS 314537 TO BUS 314575 CKT 1 /* 6SUFFOLK 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BUS 314575 /* ISLAND OPEN BUS 314590 /* ISLAND END
DVP_P1-2: LN 563	CONTINGENCY 'DVP_P1-2: LN 563' OPEN BRANCH FROM BUS 314902 TO BUS 314914 CKT 1 /* 8CARSON 500.00 - 8MDLTHAN 500.00 END
DVP_P1-2: LN 575	CONTINGENCY 'DVP_P1-2: LN 575' OPEN BRANCH FROM BUS 314911 TO BUS 314918 CKT 1 /* 8LDYSMTH 500.00 - 8NO ANNA 500.00 END
DVP_P1-2: LN 68	CONTINGENCY 'DVP_P1-2: LN 68' OPEN BRANCH FROM BUS 314527 TO BUS 314536 CKT 1 /* 3HOLLAND 115.00 - 3SUFFOLK 115.00 OPEN BRANCH FROM BUS 314527 TO BUS 314539 CKT 1 /* 3HOLLAND 115.00 - 3UNCAMP 115.00 OPEN BUS 314527 /* ISLAND END
DVP_P1-3: 3CAROLNA- 6CAROLNA	CONTINGENCY 'DVP_P1-3: 3CAROLNA-6CAROLNA' OPEN BRANCH FROM BUS 314559 TO BUS 314561 CKT 1 END

DVP_P4-2: 14852	CONTINGENCY 'DVP_P4-2: 14852' /* CLUBHOUSE OPEN BUS 314562 /* CLUBHOUSE 115KV BUS OPEN BUS 314325 /* LINE 148 OPEN BUS 314312 /* LINE 148 END
DVP_P4-2: 2012TH4	CONTINGENCY 'DVP_P4-2: 2012TH4' /* EARLEYS OPEN BRANCH FROM BUS 314569 TO BUS 936400 CKT 1 /* 2012 /* CONTINGENCY LINE ADDED FOR AD2 BUILD OPEN BRANCH FROM BUS 314569 TO BUS 314568 CKT 2 /* TX. #4 END
DVP_P4-2: 246T2034	CONTINGENCY 'DVP_P4-2: 246T2034' /* EARLEYS OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 246 OPEN BRANCH FROM BUS 314575 TO BUS 314537 CKT 1 /* 246 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 246 - NUCOR OPEN BRANCH FROM BUS 314569 TO BUS 314620 CKT 1 /* 2034 OPEN BRANCH FROM BUS 314620 TO BUS 933990 CKT 1 /* 2034 END
DVP_P4-2: 246T247	CONTINGENCY 'DVP_P4-2: 246T247' /* SUFFOLK 230 KV OPEN BRANCH FROM BUS 314537 TO BUS 314575 CKT 1 /* 6SUFFOLK 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1 /* 6EARLEYS 230.00 - 6NUCO TP 230.00 OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1 /* 6NUCO TP 230.00 - 6NUCOR 230.00 OPEN BUS 314575 /* ISLAND: 6NUCO TP 230.00 OPEN BUS 314590 /* ISLAND: 6NUCOR 230.00 OPEN BRANCH FROM BUS 314537 TO BUS 314648 CKT 1 /* 6SUFFOLK 230.00 - 6SUNBURY 230.00 OPEN BRANCH FROM BUS 314648 TO BUS 901080 CKT 1 /* 6SUNBURY 230.00 - W1-029 230.00 OPEN BUS 314648 /* ISLAND: 6SUNBURY 230.00 END
DVP_P4-2: 7152	CONTINGENCY 'DVP_P4-2: 7152' /* CLUBHOUSE OPEN BUS 314562 /* CLUBHOUSE 115KV BUS OPEN BUS 314565 /* LINE 71 OPEN BUS 314850 /* LINE 71 OPEN BUS 314693 /* LINE 71 OPEN BUS 314704 /* LINE 71 OPEN BUS 314676 /* LINE 71 OPEN BUS 314782 /* LINE 132 OPEN BUS 314786 /* LINE 132 OPEN BUS 314787 /* LINE 50 OPEN BUS 314782 /* LINE 50

AD2-090 Handsom - Southampton 115 kV

Summer Peak Analysis – 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

Contingency					Bus			Loading %			Rating		MW	Flowgate
#	Type	Name	Affected Area	Facility Description	From	To	Ckt	Power Flow	Initial	Final	Type	MVA	Contribution	Appendix
1	N-1	DVP_P1-2: LN 2056-A	DVP - DVP	3COX DP-3CHESTNUT 115 kV line	314577	313719	1	AC	98.66	100.61	ER	134	2.63	1
2	N-1	DVP_P1-2: LN 130-A	DVP - DVP	AB2-100 TAP-6CLUBHSE 230 kV line	924510	314563	1	AC	99.29	100.48	ER	375	4.61	2

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output).

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Contributions to previously identified circuit breakers found to be over-duty:

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Contingency					Bus			Power Flow	Loading %		Rating		MW Contribution	Flowgate Appendix
#	Type	Name	Affected Area	Facility Description	From	To	Ckt		Initial	Final	Type	MVA		
3	N-1	DVP_P1-2: LN 2056-A	DVP - DVP	3CHESTNUT-3WITAKRS 115 kV line	313719	314623	1	AC	131.78	133.71	ER	134	2.62	4
4	N-1	DVP_P1-2: LN 2056-B	DVP - DVP	3CHESTNUT-3WITAKRS 115 kV line	313719	314623	1	AC	129.3	131.21	ER	134	2.62	
5	DCTL	DVP_P7-1: LN 81-2056	DVP - CPLE	6MORNSTR-6ROCKYMT230T 230 kV line	313845	304222	1	AC	154.38	155.71	ER	374	6.03	5
6	DCTL	DVP_P7-1: LN 2058-2181	DVP - CPLE	3BTLEBRO-3ROCKYMT115T 115 kV line	314554	304223	1	AC	261.68	264.14	ER	164	4.79	6
7	LFFB	DVP_P4-2: 7152	DVP - DVP	6CAROLNA 230/115 kV transformer	314559	314561	1	AC	120.91	125.2		289	12.51	7
8	LFFB	DVP_P4-2: 14852	DVP - DVP	6CAROLNA 230/115 kV transformer	314559	314561	1	AC	119.77	124.06		289	12.51	
9	LFFB	DVP_P4-3: L152	DVP - DVP	6CAROLNA 230/115 kV transformer	314559	314561	1	AC	119.77	124.06		289	12.51	
10	LFFB	DVP_P4-2: 246T247	DVP - DVP	6CLUBHSE-6SAPONY 230 kV line	314563	314435	1	AC	119.67	120.92		637	9.38	8
11	LFFB	DVP_P4-2: 246T2034	DVP - DVP	6CLUBHSE-6SAPONY 230 kV line	314563	314435	1	AC	116.94	118.3		637	10.24	
12	LFFB	DVP_P4-2: 2012TH4	DVP - DVP	6EARLEYS 230/115 kV transformer	314568	314569	1	AC	102.3	106.91		208	10.34	9
13	DCTL	DVP_P7-1: LN 2058-2181	DVP - DVP	6EVERETS-AD2-068 TAP 230 kV line	314574	936530	1	AC	131.9	133.03		485	6.55	10
14	N-1	DVP_P1-2: LN 2056-A	DVP - DVP	3WITAKRS-3BTLEBRO 115 kV line	314623	314554	1	AC	132.75	134.68	ER	134	2.61	11
15	N-1	DVP_P1-2: LN 2056-B	DVP - DVP	3WITAKRS-3BTLEBRO 115 kV line	314623	314554	1	AC	130.27	132.18	ER	134	2.61	
16	N-1	DVP_P1-2: LN 246	DVP - DVP	AB2-100 TAP-6CLUBHSE 230 kV line	924510	314563	1	AC	102.11	103.09	ER	375	3.77	
17	DCTL	DVP_P7-1: LN 2058-2181	DVP - CPLE	AD2-068 TAP-6GREENVILE T 230 kV line	936530	304451	1	AC	140.08	141.23	ER	478	6.55	12

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

To be completed during the Facilities Study phase.

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be completed during the Facilities Study phase.

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this interconnection request)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation
# 1	3COX DP-3CHESTNUT 115 kV line	Description: Wreck and rebuild 0.98 miles of the Chestnut - Cox DP 115 kV line Rating: 262 MVA (normal), 287 MVA (emergency), and 349 MVA (load dump) Schedule: 20-24 months (A NC PUC approval maybe required) Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AD2-090 could become the driver and could be responsible for the upgrade. Note 2: Although Queue Project AD2-090 may not have cost responsibility for this upgrade, Queue Project AD2-090 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the upgrade, Queue Project AD2-090 will need an interim study.	n6248	\$1,960,000	\$0
# 2, 16	AB2-100 TAP-6CLUBHSE 230 kV line	Description: Rebuild Clubhouse-Lakeview 230 kV Line #254 Schedule: 12/31/2024 in-service date	b3121	\$27,000,000	\$0
Total New Network Upgrades					\$ 0

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which is calculated and reported for in the Impact Study)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation
# 3, 4	AB2-100 TAP-6CLUBHSE 230 kV line	Description: Wreck and rebuild 4.39 miles of the Chestnut – Whitakers 115 kV line #1001 Rating: 261/261/301 Schedule: 30 - 36 months Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AD2-090 could become the driver and could be responsible for the upgrade Note 2: Although Queue Project AD2-090 may not have cost responsibility for this upgrade, Queue Project AD2-090 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the upgrade, Queue Project AD2-090 will need an interim study.	n6222	\$8,780,000	\$0

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation
# 5	6MORNSTR-6ROCKYMT230T 230 kV line	Description: Rebuild the 4.3 miles of Dominion 230 kV Line #2058 Rocky Mt. – Hathaway Rating: 1047/1047/1204 Schedule: 12/31/2024 in-service date Note: Although Queue Project AD2-090 may not have cost responsibility for this upgrade, Queue Project AD2-090 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the upgrade, Queue Project AD2-090 will need an interim study.	b3122	\$13,000,000	\$0
# 6	3BTLEBRO-3ROCKYMT115T 115 kV line	A potential constraint was identified by PJM on the Battleboro – Rocky Mt. 115 kV line (Dominion – Duke Energy/Progress (DEP) tie line). There are no mitigations currently planned for the DEP portion of this overload. The Queue Project AD2-090 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Battleboro – Rocky Mt. 115 kV line prior to the execution of the Queue Project’s final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the baseline Network Upgrade, Queue Project AD2-090 will need an interim deliverability study.			\$0

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation																		
# 7, 8, 9	6CAROLNA 230/115 kV transformer	Description: Add a second 230-115 kV transformer at the Carolina substation Rating: 276.8 MVA (normal), 292.4 MVA (emergency), and 328.7 MVA (load dump). <table><tr><th>Queue</th><th>MW contribution</th><th>Cost</th></tr><tr><td>AD1-047</td><td>32.81</td><td>\$2,162,951</td></tr><tr><td>AD2-033</td><td>11.18</td><td>\$737,025</td></tr><tr><td>AD2-046</td><td>12.53</td><td>\$826,022</td></tr><tr><td>AD2-063</td><td>14.4</td><td>\$949,299</td></tr><tr><td>AD2-090</td><td>12.51</td><td>\$824,703</td></tr></table>	Queue	MW contribution	Cost	AD1-047	32.81	\$2,162,951	AD2-033	11.18	\$737,025	AD2-046	12.53	\$826,022	AD2-063	14.4	\$949,299	AD2-090	12.51	\$824,703	n6113	\$5,500,000	\$824,703
Queue	MW contribution	Cost																					
AD1-047	32.81	\$2,162,951																					
AD2-033	11.18	\$737,025																					
AD2-046	12.53	\$826,022																					
AD2-063	14.4	\$949,299																					
AD2-090	12.51	\$824,703																					
# 10, 11	6CLUBHSE-6SAPONY 230 kV line	Description: Replace wave trap at Clubhouse Substation. Rating: 722 MVA Schedule: 12 - 16 months Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AD2-090 could receive cost allocation. Note 2: Although Queue Project AD2-090 may not have cost responsibility for this upgrade, Queue Project AD2-090 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the upgrade, Queue Project AD2-090 will need an interim study.	n6051	\$150,000	\$0																		

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation									
# 12	6EARLEYS 230/115 kV transformer	Description: Add a second 230-115 kV transformer at the Earleys substation Rating: 276.8 MVA (normal), 292.4 MVA (emergency), and 328.7 MVA (load dump). <table><tr><td>Queue</td><td>MW contribution</td><td>Cost</td></tr><tr><td>AD1-047</td><td>1.71</td><td>\$779,322</td></tr><tr><td>AD2-090</td><td>10.34</td><td>\$4,720,677</td></tr></table>	Queue	MW contribution	Cost	AD1-047	1.71	\$779,322	AD2-090	10.34	\$4,720,677	n6115	\$5,500,000	\$4,720,677
Queue	MW contribution	Cost												
AD1-047	1.71	\$779,322												
AD2-090	10.34	\$4,720,677												
# 14, 15	3WITAKRS-3BTLEBRO 115 kV line	Description: Wreck and rebuild 4.55miles of the Whitakers–Battleboro 115kV line # 1001 Note 1: As changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AD2-090 could become the driver and could be responsible for the upgrade Note 2: Although Queue Project AD2-090 may not have cost responsibility for this upgrade, Queue Project AD2-090 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the upgrade, Queue Project AD2-090 will need an interim study.	n6321	\$9,100,000	\$0									

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AD2-090 Allocation																								
# 13, 17	6EVERETS-AD2-068 TAP 230 kV line AD2-068 TAP- 6GREENVILE T 230 kV line	Dominion Portion: Description: Rebuild the 20.5 miles of Dominion 230 kV Line #218 Everetts - Greenville Rating: 1047/1047/1204 Schedule: 30-36 months <table><tr><td>Queue</td><td>MW contribution</td><td>Cost</td></tr><tr><td>AD1-023</td><td>18.24</td><td>\$3,533,325</td></tr><tr><td>AD1-047</td><td>7.1</td><td>\$1,375,362</td></tr><tr><td>AD1-057</td><td>13.49</td><td>\$2,613,188</td></tr><tr><td>AD1-076</td><td>82.59</td><td>\$15,998,756</td></tr><tr><td>AD2-051</td><td>13.6</td><td>\$2,634,497</td></tr><tr><td>AD2-089</td><td>17.17</td><td>\$3,326,052</td></tr><tr><td>AD2-090</td><td>6.55</td><td>\$1,268,820</td></tr></table> Duke Energy/Progress Portion: A potential constraint was identified by PJM on the Duke Energy/Progress (DEP) portion of the Everetts - Greenville 230 kV line. There are no mitigations currently planned for the DEP portion of this overload. The Queue Project AD2-090 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Everetts - Greenville 230 kV line prior to the execution of the Queue Project’s final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AD2-090 comes into service prior to completion of the baseline Network Upgrade, Queue Project AD2-090 will need an interim deliverability study.	Queue	MW contribution	Cost	AD1-023	18.24	\$3,533,325	AD1-047	7.1	\$1,375,362	AD1-057	13.49	\$2,613,188	AD1-076	82.59	\$15,998,756	AD2-051	13.6	\$2,634,497	AD2-089	17.17	\$3,326,052	AD2-090	6.55	\$1,268,820	n6144	\$30,750,000	\$1,268,820
Queue	MW contribution	Cost																											
AD1-023	18.24	\$3,533,325																											
AD1-047	7.1	\$1,375,362																											
AD1-057	13.49	\$2,613,188																											
AD1-076	82.59	\$15,998,756																											
AD2-051	13.6	\$2,634,497																											
AD2-089	17.17	\$3,326,052																											
AD2-090	6.55	\$1,268,820																											
Total Previously Identified Upgrades					\$6,814,200																								

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The IC can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this interconnection request by addressing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

#	Contingency		Affected Area	Facility Description	Bus		Ckt	Power Flow	Loading %		Rating		MW Contribution
	Type	Name			From	To			Initial	Final	Type	MVA	
18	N-1	DVP_P1-2: LN 2056-B	DVP - DVP	3CHESTNUT-3WITAKRS 115 kV line	313719	314623	1	AC	165.95	168.63	ER	134	4.36
19	N-1	DVP_P1-2: LN 2181	DVP - CPLE	6MORNSTR-6ROCKYMT230T 230 kV line	313845	304222	1	AC	153.69	155.06	ER	374	6.2
20	N-1	DVP_P1-2: LN 2058	DVP - DVP	6MORNSTR-6NASH 230 kV line	313845	314591	1	AC	135.37	136.54	ER	449	6.32
21	N-1	DVP_P1-2: LN 2056-A	DVP - DVP	3SO JUSTICE-3COX DP 115 kV line	313858	314577	1	AC	121.29	123.53	ER	165	4.38
22	N-1	DVP_P1-2: LN 563	DVP - DVP	6CHESTF B-6BASIN 230 kV line	314287	314276	1	AC	139.54	140.46	ER	449	4.78
23	N-1	DVP_P1-2: LN 2131A	DVP - DVP	6SAPONY-6CARSON 230 kV line	314435	314282	1	AC	113.51	114.63	ER	679	8.83
24	N-1	DVP_P1-2: LN 140-C	DVP - DVP	3FRNKLN-3UNCAMP 115 kV line	314524	314539	1	AC	91.52	122.22	ER	225	69.97
25	N-1	DVP_P1-2: LN 140-C	DVP - DVP	3HOLLAND-3SUFFOLK 115 kV line	314527	314536	1	AC	80.93	106.53	ER	270	69.97
26	N-1	DVP_P1-2: LN 238	DVP - DVP	3S HAMPT-3WATKINS 115 kV line	314534	314541	1	AC	85.84	100.8	ER	225	34.31
27	N-1	DVP_P1-2: LN 140-C	DVP - DVP	3UNCAMP-3HOLLAND 115 kV line	314539	314527	1	AC	104.37	135.12	ER	225	69.97
28	N-1	DVP_P1-2: LN 68	DVP - DVP	3BOYKINS-3MURPHYS 115 kV line	314558	314589	1	AC	105.35	125.54	ER	117	24.61
29	N-1	DVP_P1-2: LN 130-A	DVP - DVP	6CAROLNA 230/115 kV transformer	314559	314561	1	AC	135.81	141.26	ER	240	12.59
30	Non	Non	DVP - DVP	6CAROLNA 230/115 kV transformer	314559	314561	1	AC	107.36	111.83	NR	227	10.43
31	N-1	DVP_P1-3: 3CAROLNA-6CAROLNA	DVP - DVP	6CLUBHSE 230/115 kV transformer	314562	314563	1	AC	145.87	148.61	ER	183	5.83
32	N-1	DVP_P1-2: LN 246	DVP - DVP	6CLUBHSE-6SAPONY 230 kV line	314563	314435	1	AC	124.32	125.64	ER	599	9.35
33	Non	Non	DVP - DVP	6CLUBHSE-6SAPONY 230 kV line	314563	314435	1	AC	105.43	106.61	NR	599	8.44
34	N-1	DVP_P1-2: LN 2131A	DVP - DVP	6EVERETS-AD2-068 TAP 230 kV line	314574	936530	1	AC	122.96	123.89	ER	449	5.19
35	Non	Non	DVP - DVP	6EVERETS-AD2-068 TAP 230 kV line	314574	936530	1	AC	101.11	101.98	NR	449	4.75
36	N-1	DVP_P1-2: LN 2056-A	DVP - DVP	3COX DP-3CHESTNUT 115 kV line	314577	313719	1	AC	141.84	144.6	ER	134	4.38
37	Non	Non	DVP - DVP	3COX DP-3CHESTNUT 115 kV line	314577	313719	1	AC	100.84	103.36	NR	123	3.65
38	N-1	DVP_P1-2: LN 2012-B	DVP - DVP	6HORNRTN-AD1-057 TAP 230 kV line	314579	934330	1	AC	114.27	115.16	ER	442	4.79
39	N-1	DVP_P1-2: LN 130-A	DVP - DVP	6LAKEVEW-AB2-100 TAP 230 kV line	314583	924510	1	AC	135.14	137.12	ER	375	7.68
40	Non	Non	DVP - DVP	6LAKEVEW-AB2-100 TAP 230 kV line	314583	924510	1	AC	105.74	106.95	NR	375	5.44
41	N-1	DVP_P1-2: LN 2058	DVP - CPLE	6NASH-6PA-RMOUNT#4 230 kV line	314591	304226	1	AC	124.46	125.57	ER	470	6.32

42	N-1	DVP_P1-2: LN 575	DVP - DVP	8NO ANNA-8SPOTSYL 500 kV line	314918	314934	1	AC	99.55	100.08	ER	3219	11.34
43	N-1	DVP_P1-2: LN 68	DVP - DVP	AA2-088 TAP-3BOYKINS 115 kV line	920040	314558	1	AC	104.34	134.37	ER	225	69.97
44	N-1	DVP_P1-2: LN 130-A	DVP - DVP	AB2-100 TAP-6CLUBHSE 230 kV line	924510	314563	1	AC	150.03	151.99	ER	375	7.68
45	Non	Non	DVP - DVP	AB2-100 TAP-6CLUBHSE 230 kV line	924510	314563	1	AC	120.65	121.86	NR	375	5.44
46	N-1	DVP_P1-3: 3CAROLNA-6CAROLNA	DVP - DVP	AC1-208 TAP-3SO JUSTICE 115 kV line	927140	313858	1	AC	92.04	94.46	ER	165	4.73
47	N-1	DVP_P1-2: LN 238	DVP - DVP	AD1-057 TAP-6MORNSTR 230 kV line	934330	313845	1	AC	116.89	118.47	ER	442	7.18
48	N-1	DVP_P1-2: LN 2131A	DVP - CPLE	AD2-068 TAP-6GREENVILE T 230 kV line	936530	304451	1	AC	120.41	121.3	ER	478	5.19

Light Load Analysis in 2021

Not required

Affected System Analysis & Mitigation

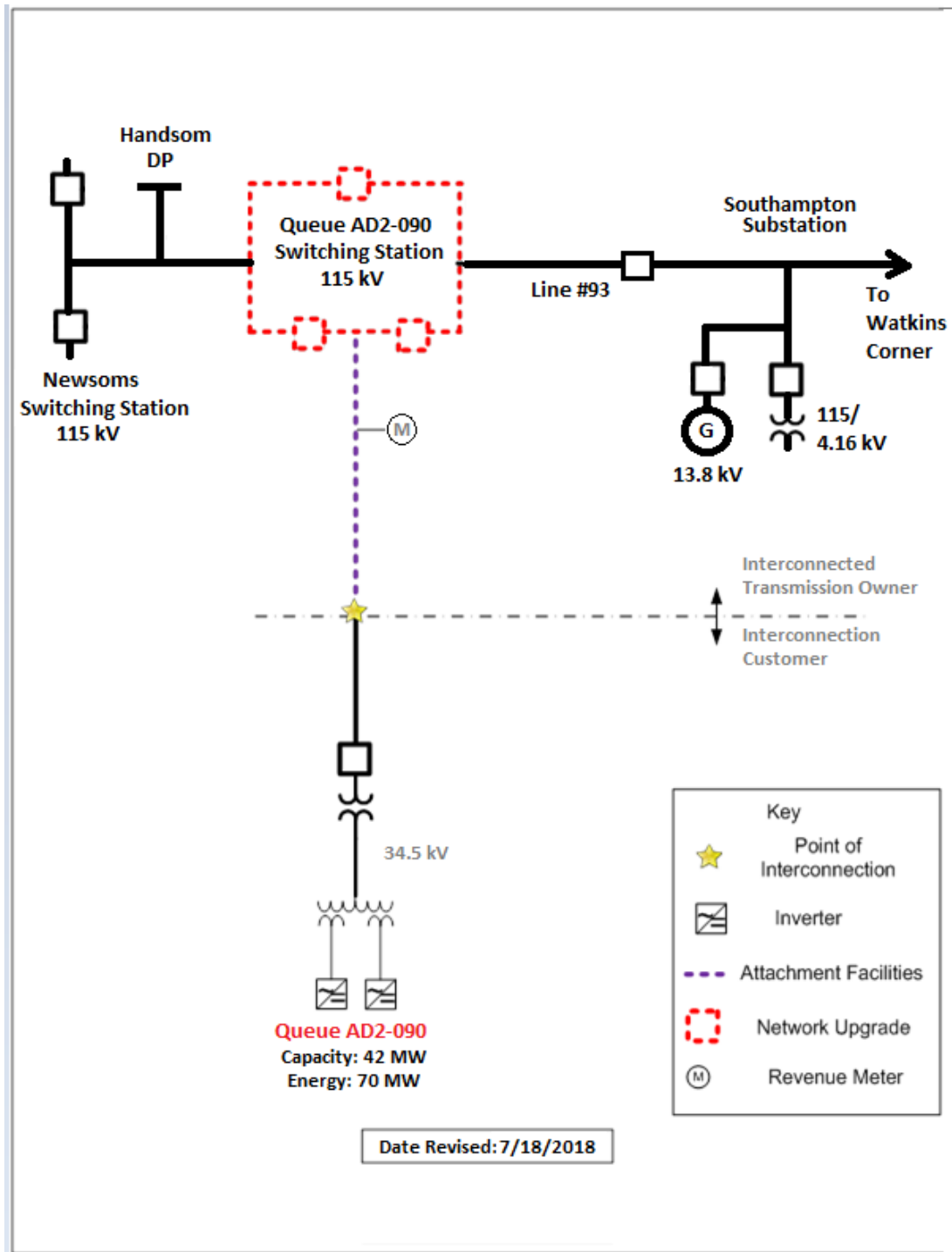
Duke Energy/Progress:

Potential constraints were identified by PJM on the following Dominion – Duke Energy/Progress (DEP) tie lines. There are no mitigations currently planned for the DEP portions of these overloads. The Queue Project AD2-090 may be subject to operational restriction if real-time system reliability issues occur. The following facilities were identified in this report:

- Battleboro – Rocky Mt. 115 kV line
- Everetts - Greenville 230 kV line
- Rocky Mt. – Hathaway 230 kV line

Attachment 1.

System Configuration



Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact.

It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

Appendix 1

(DVP - DVP) The 3COX DP-3CHESTNUT 115 kV line (from bus 314577 to bus 313719 ckt 1) loads from 98.66% to 100.61% (AC power flow) of its emergency rating (134 MVA) for the single line contingency outage of 'DVP_P1-2: LN 2056-A'. This project contributes approximately 2.63 MW to the thermal violation.

CONTINGENCY 'DVP_P1-2: LN 2056-A'

OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1

/* 6HATHAWAY

230.00 - AD1-057 TAP 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315139	1GASTONA	1.21
315141	1GASTONB	1.21
315126	1ROARAP2	1.04
315128	1ROARAP4	1.
315136	1ROSEMG1	0.87
315138	1ROSEMG2	0.41
315137	1ROSEMS1	0.54
315115	1S HAMPT1	0.6
932631	AC2-084 C	20.13
934201	AD1-047 C	3.88
934331	AD1-057 C O1	3.57
936711	AD2-090 C O1	2.63
937571	AD2-169 C	5.27
LTF	AMIL	0.14
LTF	BLUEG	0.72
LTF	CALDERWOOD	0.44
LTF	CANNELTON	0.14
LTF	CATAWBA	0.44
LTF	CBM-N	< 0.01

<i>LTF</i>	<i>CHEOAH</i>	<i>0.41</i>
<i>LTF</i>	<i>CHILHOWEE</i>	<i>0.14</i>
<i>LTF</i>	<i>CLIFTY</i>	<i>2.61</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>1.48</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.22</i>
<i>LTF</i>	<i>ELMERSMITH</i>	<i>0.41</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.18</i>
<i>LTF</i>	<i>G-007A</i>	<i>0.32</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.25</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.9</i>
<i>LTF</i>	<i>MORGAN</i>	<i>1.3</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.62</i>
<i>LTF</i>	<i>NYISO</i>	<i>0.05</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>1.34</i>
<i>LTF</i>	<i>SANTEETLA</i>	<i>0.12</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.12</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.3</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.26</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.14</i>
<i>LTF</i>	<i>TVA</i>	<i>0.55</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.75</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.05</i>
<i>LTF</i>	<i>VFT</i>	<i>0.86</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.51</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>0.92</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.08</i>

919691	AA2-053 C	1.03
920041	AA2-088 C	0.52
920671	AA2-174 C	0.05
930861	AB1-132 C	4.89
931231	AB1-173 C	1.09
931241	AB1-173AC	1.09
923801	AB2-015 C O1	2.74
923911	AB2-031 C O1	1.08
923991	AB2-040 C O1	3.56
924151	AB2-059 C O1	-10.61
924501	AB2-099 C	0.28
925171	AB2-174 C O1	3.13
925781	AC1-054 C	2.49
926071	AC1-086 C	7.2
926201	AC1-098 C	14.12
926211	AC1-099 C	4.73
927141	AC1-208 C	19.62

Appendix 2

(DVP - DVP) The AB2-100 TAP-6CLUBHSE 230 kV line (from bus 924510 to bus 314563 ckt 1) loads from 99.29% to 100.48% (AC power flow) of its emergency rating (375 MVA) for the single line contingency outage of 'DVP_P1-2: LN 130-A'. This project contributes approximately 4.61 MW to the thermal violation.

CONTINGENCY 'DVP_P1-2: LN 130-A'

```

OPEN BRANCH FROM BUS 314562 TO BUS 314570 CKT 1          /* 3CLUBHSE
115.00 - 3METCATP 115.00
OPEN BRANCH FROM BUS 314570 TO BUS 314572 CKT 1          /* 3METCATP
115.00 - 3EMPORIA 115.00
OPEN BRANCH FROM BUS 314570 TO BUS 314588 CKT 1          /* 3METCATP
115.00 - 3METCALF 115.00
OPEN BRANCH FROM BUS 314572 TO BUS 925170 CKT 1          /* 3EMPORIA
115.00 - AB2-174 TAP 115.00
OPEN BRANCH FROM BUS 314572 TO BUS 314863 CKT 1          /* 3EMPORIA
115.00 - 3EMPOR_1 115.00
OPEN BUS 314570                                           /* ISLAND
OPEN BUS 314572                                           /* ISLAND
OPEN BUS 314588                                           /* ISLAND
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315294	1DOMTR10	1.45
315292	1DOMTR78	0.98
315293	1DOMTR9	0.8
315131	1EDGECEMA	9.34
315132	1EDGECEMB	9.34
315139	1GASTONA	7.1
315141	1GASTONB	7.1
315159	1KERR 2	0.53
315163	1KERR 6	0.52
315164	1KERR 7	0.52
315126	1ROARAP2	1.97

315128	1ROARAP4	1.89
315136	1ROSEMG1	4.76
315138	1ROSEMG2	2.23
315137	1ROSEMS1	2.95
315115	1S HAMPT1	1.03
932631	AC2-084 C	8.99
933991	AD1-023 C	8.28
934201	AD1-047 C	10.05
934231	AD1-050 C	2.26
934331	AD1-057 C O1	14.7
934521	AD1-076 C O1	31.22
936401	AD2-051 C O1	7.71
936531	AD2-068 C	3.42
936701	AD2-089 C	6.68
936711	AD2-090 C O1	4.61
937571	AD2-169 C	12.57
LTF	CARR	0.09
LTF	CBM-S1	3.75
LTF	CBM-S2	7.82
LTF	CBM-W1	8.09
LTF	CBM-W2	20.09
LTF	CIN	1.81
LTF	CPL	2.68
LTF	IPL	1.15
LTF	LGEE	0.39
LTF	MEC	4.13

<i>LTF</i>	<i>MECS</i>	<i>1.79</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>0.07</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.08</i>
<i>LTF</i>	<i>WEC</i>	<i>0.5</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.42</i>
<i>917341</i>	<i>Z2-044 C</i>	<i>0.25</i>
<i>917511</i>	<i>Z2-088 C OP1</i>	<i>0.83</i>
<i>918411</i>	<i>AA1-050</i>	<i>0.7</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>1.73</i>
<i>918511</i>	<i>AA1-065 C OP</i>	<i>1.44</i>
<i>918531</i>	<i>AA1-067 C</i>	<i>0.25</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.06</i>
<i>919691</i>	<i>AA2-053 C</i>	<i>1.95</i>
<i>919701</i>	<i>AA2-057 C</i>	<i>1.36</i>
<i>LTF</i>	<i>AA2-074</i>	<i>1.83</i>
<i>920041</i>	<i>AA2-088 C</i>	<i>0.94</i>
<i>920591</i>	<i>AA2-165 C</i>	<i>0.19</i>
<i>920671</i>	<i>AA2-174 C</i>	<i>0.09</i>
<i>920691</i>	<i>AA2-178 C</i>	<i>0.83</i>
<i>930401</i>	<i>AB1-081 C</i>	<i>1.64</i>
<i>930861</i>	<i>AB1-132 C</i>	<i>28.68</i>
<i>931231</i>	<i>AB1-173 C</i>	<i>2.83</i>
<i>931241</i>	<i>AB1-173AC</i>	<i>2.83</i>
<i>923801</i>	<i>AB2-015 C O1</i>	<i>4.68</i>
<i>923911</i>	<i>AB2-031 C O1</i>	<i>2.81</i>
<i>923991</i>	<i>AB2-040 C O1</i>	<i>9.22</i>

924151	AB2-059 C O1	10.19
924401	AB2-089 C	1.03
924491	AB2-098 C	0.44
924501	AB2-099 C	0.52
924511	AB2-100 C	39.54
925121	AB2-169 C	4.29
925171	AB2-174 C O1	8.8
925591	AC1-034 C	6.6
925781	AC1-054 C	4.1
926071	AC1-086 C	42.23
926201	AC1-098 C	6.3
926211	AC1-099 C	2.11
927021	AC1-189 C	7.35
927141	AC1-208 C	10.08

Appendix 4

(DVP - DVP) The 3CHESTNUT-3WITAKRS 115 kV line (from bus 313719 to bus 314623 ckt 1) loads from 131.78% to 133.71% (AC power flow) of its emergency rating (134 MVA) for the single line contingency outage of 'DVP_P1-2: LN 2056-A'. This project contributes approximately 2.62 MW to the thermal violation.

CONTINGENCY 'DVP_P1-2: LN 2056-A'

OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1

/* 6HATHAWAY

230.00 - AD1-057 TAP 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315139	1GASTONA	1.21
315141	1GASTONB	1.21
315126	1ROARAP2	1.04
315128	1ROARAP4	1.
315136	1ROSEMG1	0.87
315138	1ROSEMG2	0.41
315137	1ROSEMS1	0.54
315115	1S HAMPT1	0.59
932631	AC2-084 C	20.11
934201	AD1-047 C	3.87
934331	AD1-057 C O1	3.55
936711	AD2-090 C O1	2.62
937571	AD2-169 C	5.26
LTF	AMIL	0.15
LTF	BLUEG	0.77
LTF	CALDERWOOD	0.46
LTF	CANNELTON	0.15
LTF	CARR	< 0.01

<i>LTF</i>	<i>CATAWBA</i>	<i>0.45</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.43</i>
<i>LTF</i>	<i>CHILHOWEE</i>	<i>0.15</i>
<i>LTF</i>	<i>CLIFTY</i>	<i>2.8</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>1.54</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.24</i>
<i>LTF</i>	<i>ELMERSMITH</i>	<i>0.43</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.19</i>
<i>LTF</i>	<i>G-007A</i>	<i>0.26</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.27</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.92</i>
<i>LTF</i>	<i>MORGAN</i>	<i>1.35</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.65</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>1.41</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>< 0.01</i>
<i>LTF</i>	<i>SANTEETLA</i>	<i>0.13</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.12</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.32</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.28</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.15</i>
<i>LTF</i>	<i>TVA</i>	<i>0.57</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.77</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.05</i>
<i>LTF</i>	<i>VFT</i>	<i>0.7</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.51</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>0.92</i>

918561	AA1-072 C	0.08
919691	AA2-053 C	1.03
919701	AA2-057 C	4.99
920041	AA2-088 C	0.52
920591	AA2-165 C	0.68
920671	AA2-174 C	0.05
930861	AB1-132 C	4.87
931231	AB1-173 C	1.09
931241	AB1-173AC	1.09
923801	AB2-015 C O1	2.73
923911	AB2-031 C O1	1.08
923991	AB2-040 C O1	3.55
924151	AB2-059 C O1	-10.63
924501	AB2-099 C	0.28
925171	AB2-174 C O1	3.12
925781	AC1-054 C	2.48
926071	AC1-086 C	7.16
926201	AC1-098 C	14.11
926211	AC1-099 C	4.73
927141	AC1-208 C	19.6

Appendix 5

(DVP - CPLE) The 6MORNSTR-6ROCKYMT230T 230 kV line (from bus 313845 to bus 304222 ckt 1) loads from 154.38% to 155.71% (AC power flow) of its emergency rating (374 MVA) for the tower line contingency outage of 'DVP_P7-1: LN 81-2056'. This project contributes approximately 6.03 MW to the thermal violation.

CONTINGENCY 'DVP_P7-1: LN 81-2056'

```

OPEN BRANCH FROM BUS 314559 TO BUS 314578 CKT 1          /* 3CAROLNA
115.00 - 3HORNRTN 115.00
OPEN BRANCH FROM BUS 314578 TO BUS 314598 CKT 1          /* 3HORNRTN
115.00 - 3ROAN DP 115.00
OPEN BRANCH FROM BUS 314598 TO BUS 314628 CKT 1          /* 3ROAN DP
115.00 - 3DARLINGT DP115.00
OPEN BUS 314578                                           /* ISLAND: 3HORNRTN 115.00
OPEN BUS 314598                                           /* ISLAND: 3ROAN DP 115.00
OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1          /* 6PA-
RMOUNT#4230.00 - 6NASH 230.00
OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1          /* 6MORNSTR
230.00 - 6NASH 230.00
OPEN BRANCH FROM BUS 304226 TO BUS 304222 CKT 1          /* 6PA-
RMOUNT#4230.00 - 6ROCKYMT230T
OPEN BUS 304226                                           /* ISLAND
OPEN BUS 314591                                           /* ISLAND: 6NASH 230.00
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315131	1EDGECEMA	24.8
315132	1EDGECEMB	24.8
315139	1GASTONA	3.87
315141	1GASTONB	3.87
315126	1ROARAP2	1.18
315128	1ROARAP4	1.14
315136	1ROSEMG1	3.24
315138	1ROSEMG2	1.52
315137	1ROSEMS1	2.01

314557	3BETHELC	1.61
314554	3BTLEBRO	1.08
314566	3CRESWEL	1.09
314572	3EMPORIA	0.27
314582	3KELFORD	0.7
314603	3SCOT NK	3.23
314617	3TUNIS	0.55
314539	3UNCAMP	1.04
314541	3WATKINS	0.33
314620	6CASHIE	0.49
314574	6EVERETS	1.81
314594	6PLYMOTH	0.44
932631	AC2-084 C	9.38
932632	AC2-084 E	4.62
933991	AD1-023 C	7.25
933992	AD1-023 E	3.95
934201	AD1-047 C	5.53
934202	AD1-047 E	3.69
934331	AD1-057 C O1	19.78
934332	AD1-057 E O1	10.55
934521	AD1-076 C O1	28.5
934522	AD1-076 E O1	14.51
936401	AD2-051 C O1	5.89
936402	AD2-051 E O1	2.53
936701	AD2-089 C	14.47
936702	AD2-089 E	9.65

936711	AD2-090 C O1	3.62
936712	AD2-090 E O1	2.41
937571	AD2-169 C	7.06
937572	AD2-169 E	4.71
LTF	AMIL	0.38
LTF	BLUEG	2.
LTF	CALDERWOOD	1.17
LTF	CANNELTON	0.38
LTF	CARR	< 0.01
LTF	CATAWBA	1.14
LTF	CELEVELAND /* 35% REVERSE 4479079 4642907	< 0.01
LTF	CHEOAH	1.09
LTF	CHILHOWEE	0.38
LTF	CLIFTY	7.36
LTF	COTTONWOOD	3.92
LTF	EDWARDS	0.62
LTF	ELMERSMITH	1.11
LTF	FARMERCITY	0.48
LTF	G-007A	0.75
LTF	GIBSON	0.7
LTF	HAMLET	2.25
LTF	MORGAN	3.44
LTF	NEWTON	1.69
LTF	PRAIRIE	3.63
LTF	RENSSELAER	< 0.01
LTF	SANTEETLA	0.32

<i>LTF</i>	<i>SMITHLAND</i>	<i>0.32</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.82</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.73</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.38</i>
<i>LTF</i>	<i>TVA</i>	<i>1.45</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>1.95</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.06</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.18</i>
<i>LTF</i>	<i>VFT</i>	<i>1.99</i>
<i>907092</i>	<i>X1-038 E</i>	<i>2.6</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.37</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>0.84</i>
<i>917341</i>	<i>Z2-044 C</i>	<i>0.33</i>
<i>917342</i>	<i>Z2-044 E</i>	<i>0.75</i>
<i>917511</i>	<i>Z2-088 C OP1</i>	<i>1.62</i>
<i>917512</i>	<i>Z2-088 E OP1</i>	<i>6.74</i>
<i>918411</i>	<i>AA1-050</i>	<i>1.36</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>1.1</i>
<i>918492</i>	<i>AA1-063AE OP</i>	<i>2.74</i>
<i>918511</i>	<i>AA1-065 C OP</i>	<i>1.12</i>
<i>918512</i>	<i>AA1-065 E OP</i>	<i>2.92</i>
<i>918531</i>	<i>AA1-067 C</i>	<i>0.24</i>
<i>918532</i>	<i>AA1-067 E</i>	<i>0.54</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.06</i>
<i>918562</i>	<i>AA1-072 E</i>	<i>0.14</i>
<i>919691</i>	<i>AA2-053 C</i>	<i>1.23</i>

919692	AA2-053 E	2.79
919701	AA2-057 C	1.66
919702	AA2-057 E	4.39
920042	AA2-088 E	5.93
920591	AA2-165 C	0.23
920592	AA2-165 E	0.58
920671	AA2-174 C	0.06
920672	AA2-174 E	0.32
920692	AA2-178 E	1.86
930401	AB1-081 C	2.75
930402	AB1-081 E	6.24
930861	AB1-132 C	15.62
930862	AB1-132 E	6.69
931231	AB1-173 C	1.56
931232	AB1-173 E	0.73
931241	AB1-173AC	1.56
931242	AB1-173AE	0.73
923801	AB2-015 C O1	3.94
923802	AB2-015 E O1	3.23
923852	AB2-025 E	0.45
923911	AB2-031 C O1	1.54
923912	AB2-031 E O1	0.76
923991	AB2-040 C O1	5.07
923992	AB2-040 E O1	4.15
924151	AB2-059 C O1	17.15
924152	AB2-059 E O1	8.83

924491	AB2-098 C	0.42
924492	AB2-098 E	0.18
924501	AB2-099 C	0.4
924502	AB2-099 E	0.17
924511	AB2-100 C	8.3
924512	AB2-100 E	4.09
925121	AB2-169 C	4.03
925122	AB2-169 E	3.62
925171	AB2-174 C O1	4.75
925172	AB2-174 E O1	4.3
925591	AC1-034 C	11.11
925592	AC1-034 E	8.38
926071	AC1-086 C	23.
926072	AC1-086 E	10.47
926201	AC1-098 C	6.58
926202	AC1-098 E	3.92
926211	AC1-099 C	2.2
926212	AC1-099 E	1.29
927021	AC1-189 C	12.21
927022	AC1-189 E	6.08
927141	AC1-208 C	10.44
927142	AC1-208 E	4.63

Appendix 6

(DVP - CPLE) The 3BTLEBRO-3ROCKYMT115T 115 kV line (from bus 314554 to bus 304223 ckt 1) loads from 261.68% to 264.14% (AC power flow) of its emergency rating (164 MVA) for the tower line contingency outage of 'DVP_P7-1: LN 2058-2181'. This project contributes approximately 4.79 MW to the thermal violation.

CONTINGENCY 'DVP_P7-1: LN 2058-2181'

OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /*

6ROCKYMT230T230.00 - 6HATHAWAY 230.00

OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00

OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-

RMOUNT#4230.00 - 6NASH 230.00

OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY

230.00 - 6NASH 230.00

OPEN BUS 314591 /* ISLAND: 6NASH 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315131	1EDGECEMA	13.41
315132	1EDGECEMB	13.41
315139	1GASTONA	2.4
315141	1GASTONB	2.4
315126	1ROARAP2	1.
315128	1ROARAP4	0.96
315136	1ROSEMG1	1.95
315138	1ROSEMG2	0.91
315137	1ROSEMS1	1.21
314557	3BETHEL C	0.88
314554	3BTLEBRO	1.95
314572	3EMPORIA	0.2
314578	3HORNRTN	2.51
314582	3KELFORD	0.68

314603	3SCOT NK	3.67
314617	3TUNIS	0.43
314541	3WATKINS	0.26
314620	6CASHIE	0.32
314574	6EVERETS	1.04
932631	AC2-084 C	11.33
932632	AC2-084 E	5.58
933991	AD1-023 C	4.7
933992	AD1-023 E	2.56
934201	AD1-047 C	4.29
934202	AD1-047 E	2.86
934331	AD1-057 C O1	11.1
934332	AD1-057 E O1	5.92
936401	AD2-051 C O1	3.9
936402	AD2-051 E O1	1.67
936701	AD2-089 C	7.88
936702	AD2-089 E	5.25
936711	AD2-090 C O1	2.87
936712	AD2-090 E O1	1.91
937571	AD2-169 C	5.6
937572	AD2-169 E	3.73
LTF	AMIL	0.26
LTF	BLUEG	1.36
LTF	CALDERWOOD	0.8
LTF	CANNELTON	0.26
LTF	CARR	< 0.01

<i>LTF</i>	<i>CATAWBA</i>	<i>0.78</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.75</i>
<i>LTF</i>	<i>CHILHOWEE</i>	<i>0.26</i>
<i>LTF</i>	<i>CLIFTY</i>	<i>5.</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>2.68</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.42</i>
<i>LTF</i>	<i>ELMERSMITH</i>	<i>0.76</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.33</i>
<i>LTF</i>	<i>G-007A</i>	<i>0.47</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.48</i>
<i>LTF</i>	<i>HAMLET</i>	<i>1.56</i>
<i>LTF</i>	<i>MORGAN</i>	<i>2.35</i>
<i>LTF</i>	<i>NEWTON</i>	<i>1.15</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>2.48</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>< 0.01</i>
<i>LTF</i>	<i>SANTEETLA</i>	<i>0.22</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.22</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.56</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.5</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.26</i>
<i>LTF</i>	<i>TVA</i>	<i>0.99</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>1.34</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.15</i>
<i>LTF</i>	<i>VFT</i>	<i>1.26</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.36</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>0.82</i>

917341	Z2-044 C	0.55
917342	Z2-044 E	1.25
917511	Z2-088 C OP1	0.88
917512	Z2-088 E OP1	3.69
918411	AA1-050	0.74
918492	AA1-063AE OP	2.28
918512	AA1-065 E OP	1.93
918532	AA1-067 E	0.31
918561	AA1-072 C	0.05
918562	AA1-072 E	0.14
919691	AA2-053 C	1.02
919692	AA2-053 E	2.32
919701	AA2-057 C	2.51
919702	AA2-057 E	6.64
920042	AA2-088 E	4.77
920591	AA2-165 C	0.34
920592	AA2-165 E	0.87
920671	AA2-174 C	0.05
920672	AA2-174 E	0.27
930401	AB1-081 C	3.79
930402	AB1-081 E	8.59
930861	AB1-132 C	9.71
930862	AB1-132 E	4.16
931231	AB1-173 C	1.21
931232	AB1-173 E	0.56
931241	AB1-173AC	1.21

931242	AB1-173AE	0.56
923801	AB2-015 C O1	3.08
923802	AB2-015 E O1	2.53
923911	AB2-031 C O1	1.2
923912	AB2-031 E O1	0.59
923991	AB2-040 C O1	3.93
923992	AB2-040 E O1	3.22
924151	AB2-059 C O1	23.61
924152	AB2-059 E O1	12.16
924491	AB2-098 C	0.24
924492	AB2-098 E	0.1
924501	AB2-099 C	0.31
924502	AB2-099 E	0.13
924511	AB2-100 C	5.31
924512	AB2-100 E	2.62
925121	AB2-169 C	2.44
925122	AB2-169 E	2.19
925171	AB2-174 C O1	3.6
925172	AB2-174 E O1	3.26
925591	AC1-034 C	15.29
925592	AC1-034 E	11.54
926071	AC1-086 C	14.29
926072	AC1-086 E	6.5
926201	AC1-098 C	7.95
926202	AC1-098 E	4.73
926211	AC1-099 C	2.66

926212	AC1-099 E	1.56
927021	AC1-189 C	6.73
927022	AC1-189 E	3.35
927141	AC1-208 C	11.27
927142	AC1-208 E	5.

Appendix 7

(DVP - DVP) The 6CAROLNA 230/115 kV transformer (from bus 314559 to bus 314561 ckt 1) loads from 120.91% to 125.2% (AC power flow) of its load dump rating (289 MVA) for the line fault with failed breaker contingency outage of 'DVP_P4-2: 7152'. This project contributes approximately 12.51 MW to the thermal violation.

CONTINGENCY 'DVP_P4-2: 7152'

OPEN BUS 314562
 OPEN BUS 314565
 OPEN BUS 314850
 OPEN BUS 314693
 OPEN BUS 314704
 OPEN BUS 314676
 OPEN BUS 314782
 OPEN BUS 314786
 OPEN BUS 314787
 OPEN BUS 314782
 OPEN BUS 314785
 OPEN BUS 314516
 OPEN BUS 314675
 END

/* CLUBHOUSE

/* CLUBHOUSE 115KV BUS
 /* LINE 71
 /* LINE 71
 /* LINE 71
 /* LINE 71
 /* LINE 71
 /* LINE 132
 /* LINE 132
 /* LINE 50
 /* LINE 50
 /* LINE 50
 /* LINE 71
 /* LINE 71

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315159	1KERR 2	1.66
315164	1KERR 7	1.64
315126	1ROARAP2	4.21
315128	1ROARAP4	4.05
315115	1S HAMPT1	1.68
314572	3EMPORIA	1.12
314578	3HORNRTN	8.17
314582	3KELFORD	0.51
314603	3SCOT NK	2.69
314617	3TUNIS	0.63
314539	3UNCAMP	1.86

314541	3WATKINS	0.65
932631	AC2-084 C	8.22
932632	AC2-084 E	4.05
934201	AD1-047 C	21.51
934202	AD1-047 E	14.34
934231	AD1-050 C	6.14
934232	AD1-050 E	3.36
936261	AD2-033 C	6.71
936262	AD2-033 E	4.48
936361	AD2-046 C O1	8.58
936362	AD2-046 E O1	3.95
936481	AD2-063 C O1	8.66
936482	AD2-063 E O1	5.77
936711	AD2-090 C O1	7.5
936712	AD2-090 E O1	5.
937571	AD2-169 C	26.89
937572	AD2-169 E	17.92
LTF	CARR	0.03
LTF	CBM-S1	0.33
LTF	CBM-S2	0.5
LTF	CBM-W1	0.75
LTF	CBM-W2	1.75
LTF	CIN	0.18
LTF	CPL	0.11
LTF	G-007	0.13
LTF	IPL	0.12

<i>LTF</i>	<i>LGEE</i>	<i>0.04</i>
<i>LTF</i>	<i>MEC</i>	<i>0.38</i>
<i>LTF</i>	<i>MECS</i>	<i>0.16</i>
<i>LTF</i>	<i>O-066</i>	<i>0.85</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>0.02</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.11</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.31</i>
<i>LTF</i>	<i>WEC</i>	<i>0.05</i>
<i>907092</i>	<i>X1-038 E</i>	<i>4.64</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>0.62</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>3.44</i>
<i>918492</i>	<i>AA1-063AE OP</i>	<i>8.58</i>
<i>918562</i>	<i>AA1-072 E</i>	<i>0.1</i>
<i>919691</i>	<i>AA2-053 C</i>	<i>3.61</i>
<i>919692</i>	<i>AA2-053 E</i>	<i>8.2</i>
<i>919702</i>	<i>AA2-057 E</i>	<i>2.35</i>
<i>920041</i>	<i>AA2-088 C</i>	<i>1.55</i>
<i>920042</i>	<i>AA2-088 E</i>	<i>13.35</i>
<i>920592</i>	<i>AA2-165 E</i>	<i>0.31</i>
<i>920671</i>	<i>AA2-174 C</i>	<i>0.17</i>
<i>920672</i>	<i>AA2-174 E</i>	<i>0.95</i>
<i>930861</i>	<i>AB1-132 C</i>	<i>-13.28</i>
<i>931231</i>	<i>AB1-173 C</i>	<i>6.05</i>
<i>931232</i>	<i>AB1-173 E</i>	<i>2.82</i>
<i>931241</i>	<i>AB1-173AC</i>	<i>6.05</i>
<i>931242</i>	<i>AB1-173AE</i>	<i>2.82</i>

923801	AB2-015 C O1	7.5
923802	AB2-015 E O1	6.15
923911	AB2-031 C O1	6.
923912	AB2-031 E O1	2.96
923991	AB2-040 C O1	19.72
923992	AB2-040 E O1	16.13
924021	AB2-043 C O1	1.97
924022	AB2-043 E O1	3.23
924161	AB2-060 C O1	5.59
924162	AB2-060 E O1	2.63
924301	AB2-077 C O1	1.26
924302	AB2-077 E O1	0.84
924311	AB2-078 C O1	1.26
924312	AB2-078 E O1	0.84
924321	AB2-079 C O1	1.26
924322	AB2-079 E O1	0.84
924401	AB2-089 C	2.79
924402	AB2-089 E	1.43
924501	AB2-099 C	0.34
924502	AB2-099 E	0.14
925171	AB2-174 C O1	18.82
925172	AB2-174 E O1	17.03
925612	AC1-036 E	0.78
925781	AC1-054 C	10.55
925782	AC1-054 E	4.86
926201	AC1-098 C	5.77

<i>926202</i>	<i>AC1-098 E</i>	<i>3.44</i>
<i>926211</i>	<i>AC1-099 C</i>	<i>1.93</i>
<i>926212</i>	<i>AC1-099 E</i>	<i>1.14</i>
<i>927141</i>	<i>AC1-208 C</i>	<i>13.68</i>
<i>927142</i>	<i>AC1-208 E</i>	<i>6.08</i>

Appendix 8

(DVP - DVP) The 6CLUBHSE-6SAPONY 230 kV line (from bus 314563 to bus 314435 ckt 1) loads from 119.67% to 120.92% (AC power flow) of its load dump rating (637 MVA) for the line fault with failed breaker contingency outage of 'DVP_P4-2: 246T247'. This project contributes approximately 9.38 MW to the thermal violation.

```

CONTINGENCY 'DVP_P4-2: 246T247'                                /* SUFFOLK 230 KV
  OPEN BRANCH FROM BUS 314537 TO BUS 314575 CKT 1              /* 6SUFFOLK
230.00 - 6NUCO TP 230.00
  OPEN BRANCH FROM BUS 314569 TO BUS 314575 CKT 1              /* 6EARLEYS
230.00 - 6NUCO TP 230.00
  OPEN BRANCH FROM BUS 314575 TO BUS 314590 CKT 1              /* 6NUCO TP
230.00 - 6NUCOR 230.00
  OPEN BUS 314575                                                /* ISLAND: 6NUCO TP 230.00
  OPEN BUS 314590                                                /* ISLAND: 6NUCOR 230.00
  OPEN BRANCH FROM BUS 314537 TO BUS 314648 CKT 1              /* 6SUFFOLK
230.00 - 6SUNBURY 230.00
  OPEN BRANCH FROM BUS 314648 TO BUS 901080 CKT 1              /* 6SUNBURY
230.00 - W1-029 230.00
  OPEN BUS 314648                                                /* ISLAND: 6SUNBURY 230.00
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315131	1EDGECEMA	10.79
315132	1EDGECEMB	10.79
315139	1GASTONA	7.3
315141	1GASTONB	7.3
315126	1ROARAP2	2.61
315128	1ROARAP4	2.51
315136	1ROSEMG1	4.93
315138	1ROSEMG2	2.31
315137	1ROSEMS1	3.06
314557	3BETHELC	0.9
314554	3BTLEBRO	0.91

314566	3CRESWEL	1.69
314572	3EMPORIA	1.04
314578	3HORNRTN	5.39
314582	3KELFORD	1.09
314704	3LAWRENC	0.79
314603	3SCOT NK	4.38
314617	3TUNIS	1.
314541	3WATKINS	0.48
314620	6CASHIE	0.87
314574	6EVERETS	2.54
314594	6PLYMOTH	0.71
932631	AC2-084 C	11.79
932632	AC2-084 E	5.81
933991	AD1-023 C	12.46
933992	AD1-023 E	6.78
934201	AD1-047 C	17.54
934202	AD1-047 E	11.7
934231	AD1-050 C	5.07
934232	AD1-050 E	2.77
934331	AD1-057 C O1	15.82
934332	AD1-057 E O1	8.44
934521	AD1-076 C O1	47.09
934522	AD1-076 E O1	23.98
936261	AD2-033 C	9.
936262	AD2-033 E	6.
936361	AD2-046 C O1	8.36

936362	AD2-046 E O1	3.85
936401	AD2-051 C O1	11.33
936402	AD2-051 E O1	4.86
936481	AD2-063 C O1	11.69
936482	AD2-063 E O1	7.79
936531	AD2-068 C	4.52
936532	AD2-068 E	2.33
936701	AD2-089 C	8.12
936702	AD2-089 E	5.41
936711	AD2-090 C O1	5.63
936712	AD2-090 E O1	3.75
937571	AD2-169 C	20.7
937572	AD2-169 E	13.8
LTF	CARR	0.12
LTF	CBM-S1	5.4
LTF	CBM-S2	10.87
LTF	CBM-W1	11.99
LTF	CBM-W2	29.04
LTF	CIN	2.69
LTF	CPL	3.68
LTF	G-007	0.78
LTF	IPL	1.72
LTF	LGEE	0.58
LTF	MEC	6.04
LTF	MECS	2.7
LTF	O-066	4.93

<i>LTF</i>	<i>RENSSELAER</i>	<i>0.1</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.11</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.33</i>
<i>LTF</i>	<i>WEC</i>	<i>0.74</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.58</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>1.31</i>
<i>917341</i>	<i>Z2-044 C</i>	<i>0.31</i>
<i>917342</i>	<i>Z2-044 E</i>	<i>0.7</i>
<i>917511</i>	<i>Z2-088 C OP1</i>	<i>1.02</i>
<i>917512</i>	<i>Z2-088 E OP1</i>	<i>4.28</i>
<i>918411</i>	<i>AA1-050</i>	<i>0.86</i>
<i>918491</i>	<i>AA1-063AC OP</i>	<i>2.26</i>
<i>918492</i>	<i>AA1-063AE OP</i>	<i>5.64</i>
<i>918511</i>	<i>AA1-065 C OP</i>	<i>2.15</i>
<i>918512</i>	<i>AA1-065 E OP</i>	<i>5.61</i>
<i>918531</i>	<i>AA1-067 C</i>	<i>0.34</i>
<i>918532</i>	<i>AA1-067 E</i>	<i>0.76</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.09</i>
<i>918562</i>	<i>AA1-072 E</i>	<i>0.22</i>
<i>919691</i>	<i>AA2-053 C</i>	<i>2.62</i>
<i>919692</i>	<i>AA2-053 E</i>	<i>5.94</i>
<i>919701</i>	<i>AA2-057 C</i>	<i>1.71</i>
<i>919702</i>	<i>AA2-057 E</i>	<i>4.51</i>
<i>LTF</i>	<i>AA2-074</i>	<i>2.5</i>
<i>920041</i>	<i>AA2-088 C</i>	<i>1.19</i>
<i>920042</i>	<i>AA2-088 E</i>	<i>10.28</i>

920591	AA2-165 C	0.23
920592	AA2-165 E	0.59
920671	AA2-174 C	0.12
920672	AA2-174 E	0.69
920692	AA2-178 E	2.89
930401	AB1-081 C	1.94
930402	AB1-081 E	4.39
930861	AB1-132 C	29.48
930862	AB1-132 E	12.64
931231	AB1-173 C	4.93
931232	AB1-173 E	2.3
931241	AB1-173AC	4.93
931242	AB1-173AE	2.3
923801	AB2-015 C O1	5.46
923802	AB2-015 E O1	4.48
923911	AB2-031 C O1	4.9
923912	AB2-031 E O1	2.41
923991	AB2-040 C O1	16.08
923992	AB2-040 E O1	13.16
924021	AB2-043 C O1	2.67
924022	AB2-043 E O1	4.38
924151	AB2-059 C O1	12.06
924152	AB2-059 E O1	6.21
924161	AB2-060 C O1	7.58
924162	AB2-060 E O1	3.57
924301	AB2-077 C O1	1.67

924302	AB2-077 E O1	1.12
924311	AB2-078 C O1	1.67
924312	AB2-078 E O1	1.12
924321	AB2-079 C O1	1.67
924322	AB2-079 E O1	1.12
924401	AB2-089 C	2.3
924402	AB2-089 E	1.19
924491	AB2-098 C	0.59
924492	AB2-098 E	0.25
924501	AB2-099 C	0.73
924502	AB2-099 E	0.31
924511	AB2-100 C	35.87
924512	AB2-100 E	17.67
925121	AB2-169 C	6.13
925122	AB2-169 E	5.5
925171	AB2-174 C O1	16.14
925172	AB2-174 E O1	14.61
925591	AC1-034 C	7.81
925592	AC1-034 E	5.89
925612	AC1-036 E	1.04
925781	AC1-054 C	8.26
925782	AC1-054 E	3.81
926071	AC1-086 C	43.42
926072	AC1-086 E	19.76
926201	AC1-098 C	8.27
926202	AC1-098 E	4.93

<i>926211</i>	<i>AC1-099 C</i>	<i>2.77</i>
<i>926212</i>	<i>AC1-099 E</i>	<i>1.63</i>
<i>927021</i>	<i>AC1-189 C</i>	<i>9.37</i>
<i>927022</i>	<i>AC1-189 E</i>	<i>4.67</i>
<i>927141</i>	<i>AC1-208 C</i>	<i>13.09</i>
<i>927142</i>	<i>AC1-208 E</i>	<i>5.81</i>

Appendix 9

(DVP - DVP) The 6EARLEYS 230/115 kV transformer (from bus 314568 to bus 314569 ckt 1) loads from 102.3% to 106.91% (AC power flow) of its load dump rating (208 MVA) for the line fault with failed breaker contingency outage of 'DVP_P4-2: 2012TH4'. This project contributes approximately 10.34 MW to the thermal violation.

```
CONTINGENCY 'DVP_P4-2: 2012TH4'                                /* EARLEYS
  OPEN BRANCH FROM BUS 314569 TO BUS 936400 CKT 1              /* 2012 /*
CONTINGENCY LINE ADDED FOR AD2 BUILD
  OPEN BRANCH FROM BUS 314569 TO BUS 314568 CKT 2              /* TX. #4
END
```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315126	1ROARAP2	1.26
315128	1ROARAP4	1.21
315115	1S HAMPT1	1.37
314572	3EMPORIA	0.18
314578	3HORNRTN	2.79
314582	3KELFORD	2.21
314603	3SCOT NK	6.37
314617	3TUNIS	2.13
314539	3UNCAMP	1.47
314541	3WATKINS	0.53
932631	AC2-084 C	13.13
932632	AC2-084 E	6.47
934201	AD1-047 C	4.58
934202	AD1-047 E	3.05
934231	AD1-050 C	1.95
934232	AD1-050 E	1.07
936401	AD2-051 C O1	3.38

936402	AD2-051 E O1	1.45
936711	AD2-090 C O1	6.2
936712	AD2-090 E O1	4.13
937571	AD2-169 C	6.27
937572	AD2-169 E	4.18
LTF	CARR	0.02
LTF	CBM-S1	0.31
LTF	CBM-S2	0.52
LTF	CBM-W1	0.66
LTF	CBM-W2	1.63
LTF	CIN	0.15
LTF	CPL	0.12
LTF	G-007	0.09
LTF	IPL	0.1
LTF	LGEE	0.03
LTF	MEC	0.34
LTF	MECS	0.14
LTF	O-066	0.55
LTF	RENSSELAER	0.01
900671	V4-068 C	0.18
900672	V4-068 E	0.53
LTF	WEC	0.04
907092	X1-038 E	3.69
917331	Z2-043 C	1.17
917332	Z2-043 E	2.65
917342	Z2-044 E	0.31

918491	AA1-063AC OP	1.49
918492	AA1-063AE OP	3.71
918561	AA1-072 C	0.18
918562	AA1-072 E	0.44
919691	AA2-053 C	2.27
919692	AA2-053 E	5.15
919701	AA2-057 C	1.01
919702	AA2-057 E	2.66
920041	AA2-088 C	1.3
920042	AA2-088 E	11.25
920591	AA2-165 C	0.14
920592	AA2-165 E	0.35
920671	AA2-174 C	0.1
920672	AA2-174 E	0.6
930861	AB1-132 C	5.14
930862	AB1-132 E	2.2
931231	AB1-173 C	1.29
931232	AB1-173 E	0.6
931241	AB1-173AC	1.29
931242	AB1-173AE	0.6
923801	AB2-015 C O1	6.07
923802	AB2-015 E O1	4.97
923911	AB2-031 C O1	1.28
923912	AB2-031 E O1	0.63
923991	AB2-040 C O1	4.2
923992	AB2-040 E O1	3.43

924401	AB2-089 C	0.89
924402	AB2-089 E	0.46
924501	AB2-099 C	1.74
924502	AB2-099 E	0.75
925121	AB2-169 C	-2.59
925171	AB2-174 C O1	3.65
925172	AB2-174 E O1	3.31
925781	AC1-054 C	3.31
925782	AC1-054 E	1.53
926071	AC1-086 C	7.57
926072	AC1-086 E	3.44
926201	AC1-098 C	9.21
926202	AC1-098 E	5.49
926211	AC1-099 C	3.09
926212	AC1-099 E	1.81
927141	AC1-208 C	8.93
927142	AC1-208 E	3.96

Appendix 10

(DVP - DVP) The 6EVERETS-AD2-068 TAP 230 kV line (from bus 314574 to bus 936530 ckt 1) loads from 131.9% to 133.03% (AC power flow) of its load dump rating (485 MVA) for the tower line contingency outage of 'DVP_P7-1: LN 2058-2181'. This project contributes approximately 6.55 MW to the thermal violation.

CONTINGENCY 'DVP_P7-1: LN 2058-2181'

OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /*

6ROCKYMT230T230.00 - 6HATHAWAY 230.00

OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00

OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00

OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00

OPEN BUS 314591 /* ISLAND: 6NASH 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315294	1DOMTR10	3.01
315292	1DOMTR78	2.03
315293	1DOMTR9	1.66
315131	1EDGECEMA	9.28
315132	1EDGECEMB	9.28
315136	1ROSEMG1	1.91
315138	1ROSEMG2	0.9
315137	1ROSEMS1	1.19
314557	3BETHEL C	1.14
314554	3BTLEBRO	0.43
314566	3CRESWEL	2.04
314572	3EMPORIA	0.21
314578	3HORNRTN	2.04
314582	3KELFORD	0.72

314603	3SCOT NK	2.51
314617	3TUNIS	0.7
314539	3UNCAMP	1.18
314541	3WATKINS	0.36
314620	6CASHIE	0.88
314574	6EVERETS	5.39
314594	6PLYMOTH	0.83
314648	6SUNBURY	0.4
314651	6WINFALL	0.97
932631	AC2-084 C	6.17
932632	AC2-084 E	3.04
933991	AD1-023 C	13.46
933992	AD1-023 E	7.33
934201	AD1-047 C	4.29
934202	AD1-047 E	2.86
934331	AD1-057 C O1	8.8
934332	AD1-057 E O1	4.7
934521	AD1-076 C O1	54.75
934522	AD1-076 E O1	27.88
936401	AD2-051 C O1	9.51
936402	AD2-051 E O1	4.09
936701	AD2-089 C	10.3
936702	AD2-089 E	6.87
936711	AD2-090 C O1	3.93
936712	AD2-090 E O1	2.62
937571	AD2-169 C	5.53

937572	AD2-169 E	3.69
LTF	AMIL	0.49
LTF	BLUEG	2.51
LTF	CALDERWOOD	1.54
LTF	CANNELTON	0.48
LTF	CATAWBA	1.51
LTF	CBM-N	< 0.01
LTF	CELEVELAND /* 35% REVERSE 4479079 4642907	< 0.01
LTF	CHEOAH	1.44
LTF	CHILHOWEE	0.5
LTF	CLIFTY	9.06
LTF	COTTONWOOD	5.21
LTF	EDWARDS	0.78
LTF	ELMERSMITH	1.42
LTF	FARMERCITY	0.62
LTF	G-007A	1.03
LTF	GIBSON	0.88
LTF	HAMLET	3.22
LTF	MORGAN	4.57
LTF	NEWTON	2.15
LTF	NYISO	0.08
LTF	PRAIRIE	4.69
LTF	SANTEETLA	0.43
LTF	SMITHLAND	0.42
LTF	TATANKA	1.05
LTF	TILTON	0.92

<i>LTF</i>	<i>TRIMBLE</i>	<i>0.48</i>
<i>LTF</i>	<i>TVA</i>	<i>1.92</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>2.58</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.21</i>
<i>LTF</i>	<i>VFT</i>	<i>2.73</i>
<i>901082</i>	<i>W1-029E</i>	<i>23.38</i>
<i>907092</i>	<i>X1-038 E</i>	<i>2.96</i>
<i>913392</i>	<i>Y1-086 E</i>	<i>1.05</i>
<i>916042</i>	<i>Z1-036 E</i>	<i>29.12</i>
<i>917122</i>	<i>Z2-027 E</i>	<i>0.51</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.38</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>0.86</i>
<i>917342</i>	<i>Z2-044 E</i>	<i>0.33</i>
<i>917511</i>	<i>Z2-088 C OP1</i>	<i>1.47</i>
<i>917512</i>	<i>Z2-088 E OP1</i>	<i>6.13</i>
<i>918411</i>	<i>AA1-050</i>	<i>1.24</i>
<i>918492</i>	<i>AA1-063AE OP</i>	<i>2.44</i>
<i>918511</i>	<i>AA1-065 C OP</i>	<i>1.86</i>
<i>918512</i>	<i>AA1-065 E OP</i>	<i>4.84</i>
<i>918531</i>	<i>AA1-067 C</i>	<i>0.71</i>
<i>918532</i>	<i>AA1-067 E</i>	<i>1.62</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.06</i>
<i>918562</i>	<i>AA1-072 E</i>	<i>0.14</i>
<i>919692</i>	<i>AA2-053 E</i>	<i>2.58</i>
<i>919702</i>	<i>AA2-057 E</i>	<i>2.12</i>
<i>920042</i>	<i>AA2-088 E</i>	<i>6.25</i>

920592	AA2-165 E	0.28
920672	AA2-174 E	0.3
920691	AA2-178 C	1.54
920692	AA2-178 E	3.5
930402	AB1-081 E	2.42
930861	AB1-132 C	10.36
930862	AB1-132 E	4.44
931231	AB1-173 C	1.21
931232	AB1-173 E	0.56
931241	AB1-173AC	1.21
931242	AB1-173AE	0.56
923801	AB2-015 C O1	4.4
923802	AB2-015 E O1	3.61
923831	AB2-022 C	1.02
923832	AB2-022 E	0.55
923911	AB2-031 C O1	1.2
923912	AB2-031 E O1	0.59
923991	AB2-040 C O1	3.93
923992	AB2-040 E O1	3.22
924151	AB2-059 C O1	6.64
924152	AB2-059 E O1	3.42
924491	AB2-098 C	1.26
924492	AB2-098 E	0.54
924501	AB2-099 C	0.53
924502	AB2-099 E	0.23
924511	AB2-100 C	5.85

924512	<i>AB2-100 E</i>	2.88
925121	<i>AB2-169 C</i>	10.02
925122	<i>AB2-169 E</i>	8.99
925171	<i>AB2-174 C O1</i>	3.64
925172	<i>AB2-174 E O1</i>	3.29
925591	<i>AC1-034 C</i>	4.3
925592	<i>AC1-034 E</i>	3.25
926071	<i>AC1-086 C</i>	15.25
926072	<i>AC1-086 E</i>	6.94
926201	<i>AC1-098 C</i>	4.33
926202	<i>AC1-098 E</i>	2.58
926211	<i>AC1-099 C</i>	1.45
926212	<i>AC1-099 E</i>	0.85
<i>LTF</i>	<i>AC1-131</i>	5.64
927021	<i>AC1-189 C</i>	15.45
927022	<i>AC1-189 E</i>	7.7
927141	<i>AC1-208 C</i>	5.74
927142	<i>AC1-208 E</i>	2.55

Appendix 11

(DVP - DVP) The 3WITAKRS-3BTLEBRO 115 kV line (from bus 314623 to bus 314554 ckt 1) loads from 132.75% to 134.68% (AC power flow) of its emergency rating (134 MVA) for the single line contingency outage of 'DVP_P1-2: LN 2056-A'. This project contributes approximately 2.61 MW to the thermal violation.

CONTINGENCY 'DVP_P1-2: LN 2056-A'

OPEN BRANCH FROM BUS 313845 TO BUS 934330 CKT 1

/* 6HATHAWAY

230.00 - AD1-057 TAP 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315139	1GASTONA	1.2
315141	1GASTONB	1.2
315126	1ROARAP2	1.04
315128	1ROARAP4	1.
315136	1ROSEMG1	0.86
315138	1ROSEMG2	0.41
315137	1ROSEMS1	0.54
315115	1S HAMPT1	0.59
932631	AC2-084 C	20.11
934201	AD1-047 C	3.87
934331	AD1-057 C O1	3.55
936711	AD2-090 C O1	2.61
937571	AD2-169 C	5.25
LTF	AMIL	0.15
LTF	BLUEG	0.77
LTF	CALDERWOOD	0.46
LTF	CANNELTON	0.15
LTF	CARR	< 0.01

<i>LTF</i>	<i>CATAWBA</i>	<i>0.45</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.43</i>
<i>LTF</i>	<i>CHILHOWEE</i>	<i>0.15</i>
<i>LTF</i>	<i>CLIFTY</i>	<i>2.83</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>1.55</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.24</i>
<i>LTF</i>	<i>ELMERSMITH</i>	<i>0.43</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.19</i>
<i>LTF</i>	<i>G-007A</i>	<i>0.25</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.27</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.92</i>
<i>LTF</i>	<i>MORGAN</i>	<i>1.36</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.66</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>1.42</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>< 0.01</i>
<i>LTF</i>	<i>SANTEETLA</i>	<i>0.13</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.13</i>
<i>LTF</i>	<i>TATANKA</i>	<i>0.32</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.28</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.15</i>
<i>LTF</i>	<i>TVA</i>	<i>0.57</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.77</i>
<i>900671</i>	<i>V4-068 C</i>	<i>0.05</i>
<i>LTF</i>	<i>VFT</i>	<i>0.67</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.51</i>
<i>917341</i>	<i>Z2-044 C</i>	<i>1.13</i>

918491	AA1-063AC OP	0.92
918561	AA1-072 C	0.08
919691	AA2-053 C	1.03
919701	AA2-057 C	4.99
920041	AA2-088 C	0.52
920591	AA2-165 C	0.68
920671	AA2-174 C	0.05
930861	AB1-132 C	4.86
931231	AB1-173 C	1.09
931241	AB1-173AC	1.09
923801	AB2-015 C O1	2.72
923911	AB2-031 C O1	1.08
923991	AB2-040 C O1	3.54
924151	AB2-059 C O1	-10.63
924501	AB2-099 C	0.28
925171	AB2-174 C O1	3.11
925781	AC1-054 C	2.48
926071	AC1-086 C	7.16
926201	AC1-098 C	14.11
926211	AC1-099 C	4.73
927141	AC1-208 C	19.6

Appendix 12

(DVP - CPLE) The AD2-068 TAP-6GREENVILE T 230 kV line (from bus 936530 to bus 304451 ckt 1) loads from 140.08% to 141.23% (AC power flow) of its emergency rating (478 MVA) for the tower line contingency outage of 'DVP_P7-1: LN 2058-2181'. This project contributes approximately 6.55 MW to the thermal violation.

CONTINGENCY 'DVP_P7-1: LN 2058-2181'

OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /*

6ROCKYMT230T230.00 - 6HATHAWAY 230.00

OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00

OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-
RMOUNT#4230.00 - 6NASH 230.00

OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY
230.00 - 6NASH 230.00

OPEN BUS 314591 /* ISLAND: 6NASH 230.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
315294	1DOMTR10	3.01
315292	1DOMTR78	2.03
315293	1DOMTR9	1.66
315131	1EDGECEMA	9.28
315132	1EDGECEMB	9.28
315136	1ROSEMG1	1.91
315138	1ROSEMG2	0.9
315137	1ROSEMS1	1.19
314557	3BETHEL C	1.14
314554	3BTLEBRO	0.43
314566	3CRESWEL	2.04
314572	3EMPORIA	0.21
314578	3HORNRTN	2.04
314582	3KELFORD	0.72

314603	3SCOT NK	2.51
314617	3TUNIS	0.7
314539	3UNCAMP	1.18
314541	3WATKINS	0.36
314620	6CASHIE	0.88
314574	6EVERETS	5.39
314594	6PLYMOTH	0.83
314648	6SUNBURY	0.4
314651	6WINFALL	0.97
932631	AC2-084 C	6.17
932632	AC2-084 E	3.04
933991	AD1-023 C	13.46
933992	AD1-023 E	7.33
934201	AD1-047 C	4.29
934202	AD1-047 E	2.86
934331	AD1-057 C O1	8.8
934332	AD1-057 E O1	4.7
934521	AD1-076 C O1	54.75
934522	AD1-076 E O1	27.88
936401	AD2-051 C O1	9.51
936402	AD2-051 E O1	4.09
936531	AD2-068 C	19.96
936532	AD2-068 E	10.28
936701	AD2-089 C	10.3
936702	AD2-089 E	6.87
936711	AD2-090 C O1	3.93

936712	AD2-090 E O1	2.62
937571	AD2-169 C	5.53
937572	AD2-169 E	3.69
LTF	AMIL	0.49
LTF	BLUEG	2.51
LTF	CALDERWOOD	1.54
LTF	CANNELTON	0.48
LTF	CATAWBA	1.51
LTF	CBM-N	< 0.01
LTF	CELEVELAND /* 35% REVERSE 4479079 4642907	< 0.01
LTF	CHEOAH	1.44
LTF	CHILHOWEE	0.5
LTF	CLIFTY	9.06
LTF	COTTONWOOD	5.21
LTF	EDWARDS	0.78
LTF	ELMERSMITH	1.42
LTF	FARMERCITY	0.62
LTF	G-007A	1.03
LTF	GIBSON	0.88
LTF	HAMLET	3.22
LTF	MORGAN	4.57
LTF	NEWTON	2.15
LTF	NYISO	0.08
LTF	PRAIRIE	4.69
LTF	SANTEETLA	0.43
LTF	SMITHLAND	0.42

<i>LTF</i>	<i>TATANKA</i>	<i>1.05</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.92</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.48</i>
<i>LTF</i>	<i>TVA</i>	<i>1.92</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>2.58</i>
<i>900672</i>	<i>V4-068 E</i>	<i>0.21</i>
<i>LTF</i>	<i>VFT</i>	<i>2.73</i>
<i>901082</i>	<i>W1-029E</i>	<i>23.38</i>
<i>907092</i>	<i>X1-038 E</i>	<i>2.96</i>
<i>913392</i>	<i>Y1-086 E</i>	<i>1.05</i>
<i>916042</i>	<i>Z1-036 E</i>	<i>29.12</i>
<i>917122</i>	<i>Z2-027 E</i>	<i>0.51</i>
<i>917331</i>	<i>Z2-043 C</i>	<i>0.38</i>
<i>917332</i>	<i>Z2-043 E</i>	<i>0.86</i>
<i>917342</i>	<i>Z2-044 E</i>	<i>0.33</i>
<i>917511</i>	<i>Z2-088 C OP1</i>	<i>1.47</i>
<i>917512</i>	<i>Z2-088 E OP1</i>	<i>6.13</i>
<i>918411</i>	<i>AA1-050</i>	<i>1.24</i>
<i>918492</i>	<i>AA1-063AE OP</i>	<i>2.44</i>
<i>918511</i>	<i>AA1-065 C OP</i>	<i>1.86</i>
<i>918512</i>	<i>AA1-065 E OP</i>	<i>4.84</i>
<i>918531</i>	<i>AA1-067 C</i>	<i>0.71</i>
<i>918532</i>	<i>AA1-067 E</i>	<i>1.62</i>
<i>918561</i>	<i>AA1-072 C</i>	<i>0.06</i>
<i>918562</i>	<i>AA1-072 E</i>	<i>0.14</i>
<i>919692</i>	<i>AA2-053 E</i>	<i>2.58</i>

919702	AA2-057 E	2.12
920042	AA2-088 E	6.25
920592	AA2-165 E	0.28
920672	AA2-174 E	0.3
920691	AA2-178 C	1.54
920692	AA2-178 E	3.5
930402	AB1-081 E	2.42
930861	AB1-132 C	10.36
930862	AB1-132 E	4.44
931231	AB1-173 C	1.21
931232	AB1-173 E	0.56
931241	AB1-173AC	1.21
931242	AB1-173AE	0.56
923801	AB2-015 C O1	4.4
923802	AB2-015 E O1	3.61
923831	AB2-022 C	1.02
923832	AB2-022 E	0.55
923911	AB2-031 C O1	1.2
923912	AB2-031 E O1	0.59
923991	AB2-040 C O1	3.93
923992	AB2-040 E O1	3.22
924151	AB2-059 C O1	6.64
924152	AB2-059 E O1	3.42
924491	AB2-098 C	1.26
924492	AB2-098 E	0.54
924501	AB2-099 C	0.53

924502	<i>AB2-099 E</i>	<i>0.23</i>
924511	<i>AB2-100 C</i>	<i>5.85</i>
924512	<i>AB2-100 E</i>	<i>2.88</i>
925121	<i>AB2-169 C</i>	<i>10.02</i>
925122	<i>AB2-169 E</i>	<i>8.99</i>
925171	<i>AB2-174 C O1</i>	<i>3.64</i>
925172	<i>AB2-174 E O1</i>	<i>3.29</i>
925591	<i>AC1-034 C</i>	<i>4.3</i>
925592	<i>AC1-034 E</i>	<i>3.25</i>
926071	<i>AC1-086 C</i>	<i>15.25</i>
926072	<i>AC1-086 E</i>	<i>6.94</i>
926201	<i>AC1-098 C</i>	<i>4.33</i>
926202	<i>AC1-098 E</i>	<i>2.58</i>
926211	<i>AC1-099 C</i>	<i>1.45</i>
926212	<i>AC1-099 E</i>	<i>0.85</i>
<i>LTF</i>	<i>AC1-131</i>	<i>5.64</i>
927021	<i>AC1-189 C</i>	<i>15.45</i>
927022	<i>AC1-189 E</i>	<i>7.7</i>
927141	<i>AC1-208 C</i>	<i>5.74</i>
927142	<i>AC1-208 E</i>	<i>2.55</i>